

Logistic Population growth

Beth Babcock

2025-08-28

Open ggplot

```
library(ggplot2)
theme_set(theme_bw(base_size = 14))
```

How to model density dependence and carrying capacity?

Define K , carrying capacity as the number of animals a habitat can support.

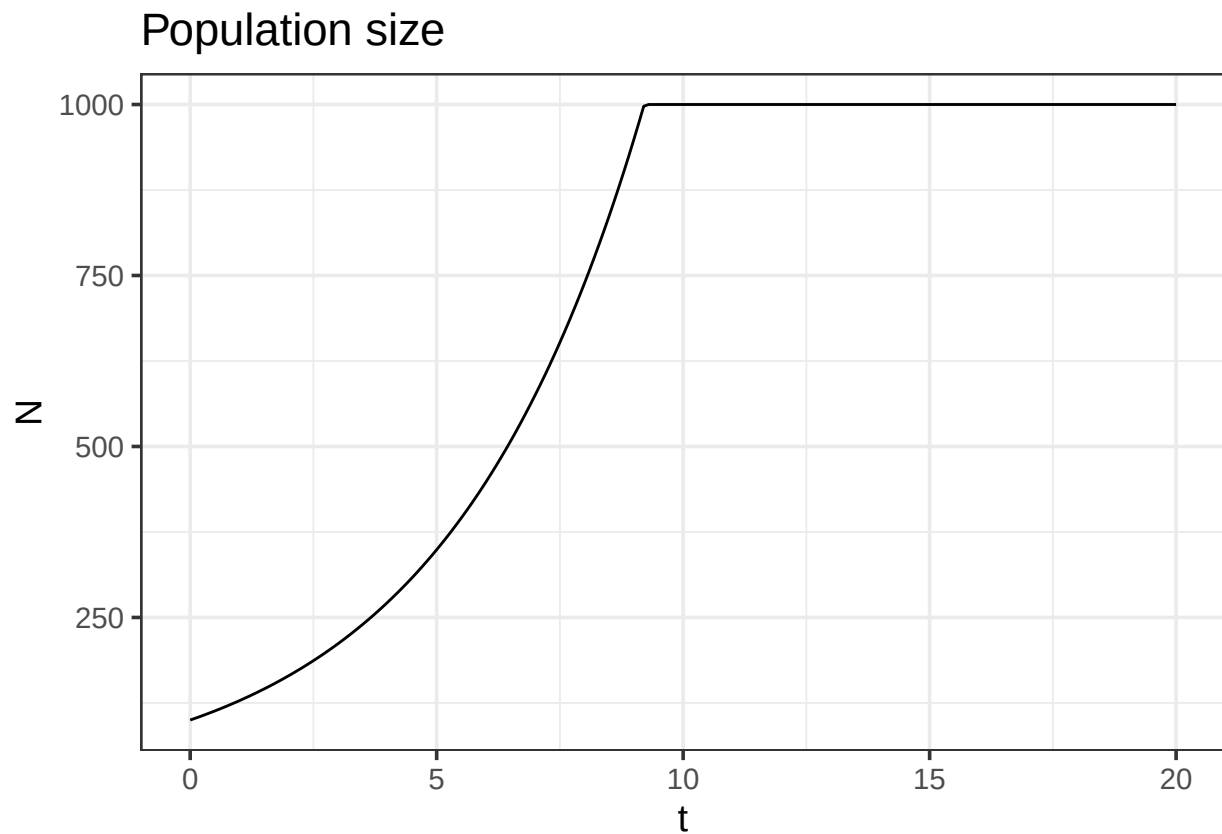
Define r_m as the intrinsic population growth rate. The actual growth rate r may be lower due to density dependent processes.

Oversimplified idea 1.

Assume that the growth rate does not change with density, but all animals after animal K die.

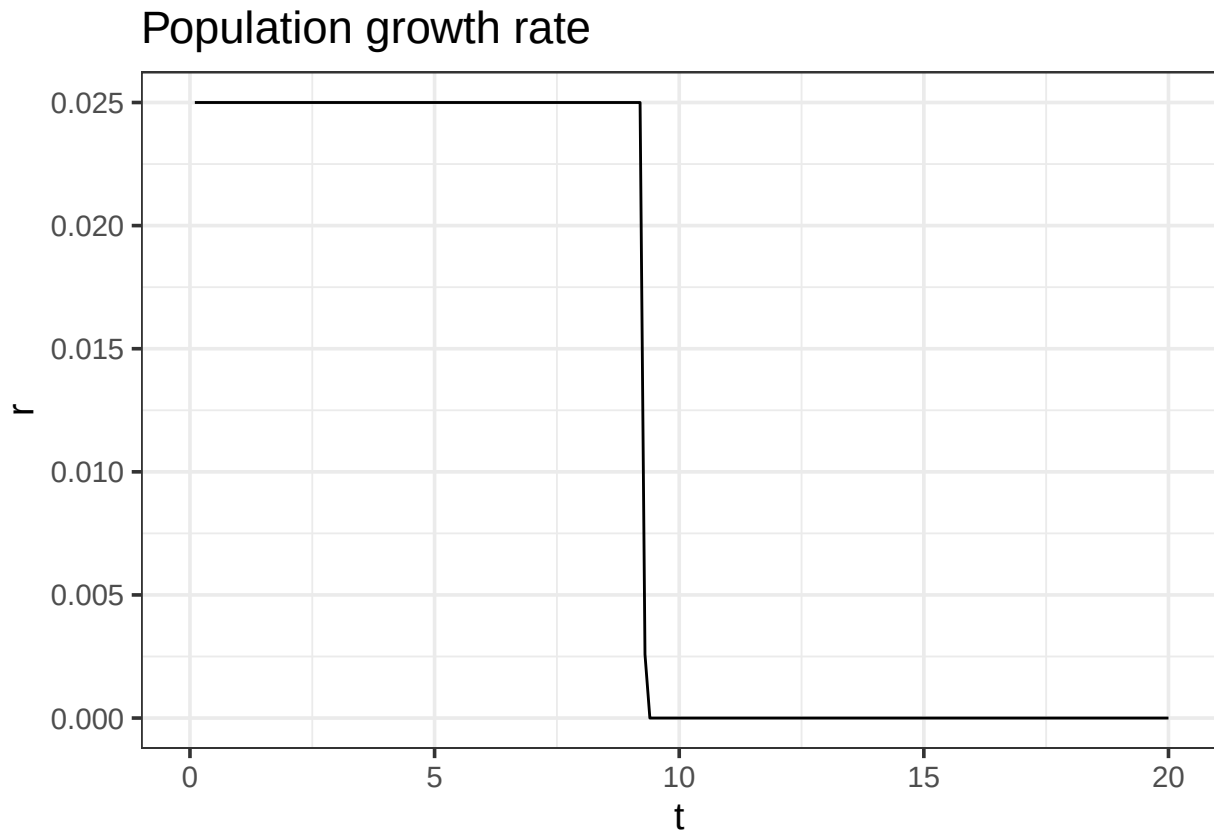
```
rm<-0.25
K<-1000
N0<-100
Population<-data.frame(t=seq(0,20,0.1))
Population$N<-pmin(N0*exp(rm*Population$t),K)

ggplot(Population,
       aes(x=t,y=N))+
  geom_line()+
  ggtitle("Population size")
```



```
Population$r<-c(NA,log(Population$N[2:nrow(Population)]/Population$N[1:(nrow(Population)-1)]))
ggplot(Population,
       aes(x=t,y=r))+
  geom_line()+
  ggtitle("Population growth rate")
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_line()').
```

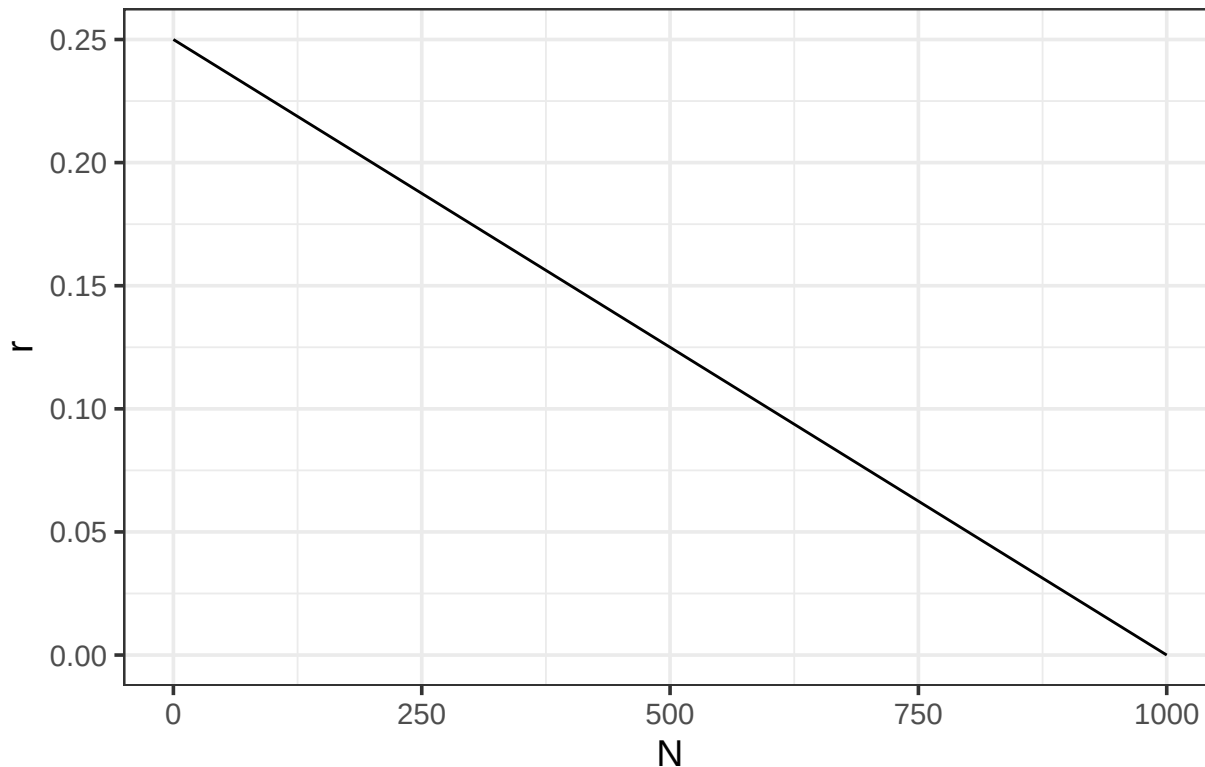


Oversimplified idea 2.

Realized r decreases linearly with abundance

```
ggplot(data.frame(N=0:1000,  
  r=(1-(0:1000)/1000)*rm),  
  aes(x=N,y=r))+  
  geom_line()+  
  ggtitle("Linear decrease in r with increasing population")
```

Linear decrease in r with increasing population



$$r = r_m(1 - N/K)$$

so that

$$\delta N / \delta t = r_m(1 - N/K)N$$

In continuous time, this integrates to:

$$N_t = \frac{K}{1 + \frac{(K-N_0)}{N_0} e^{-r_m t}}$$

We can make a function for ease of calculation

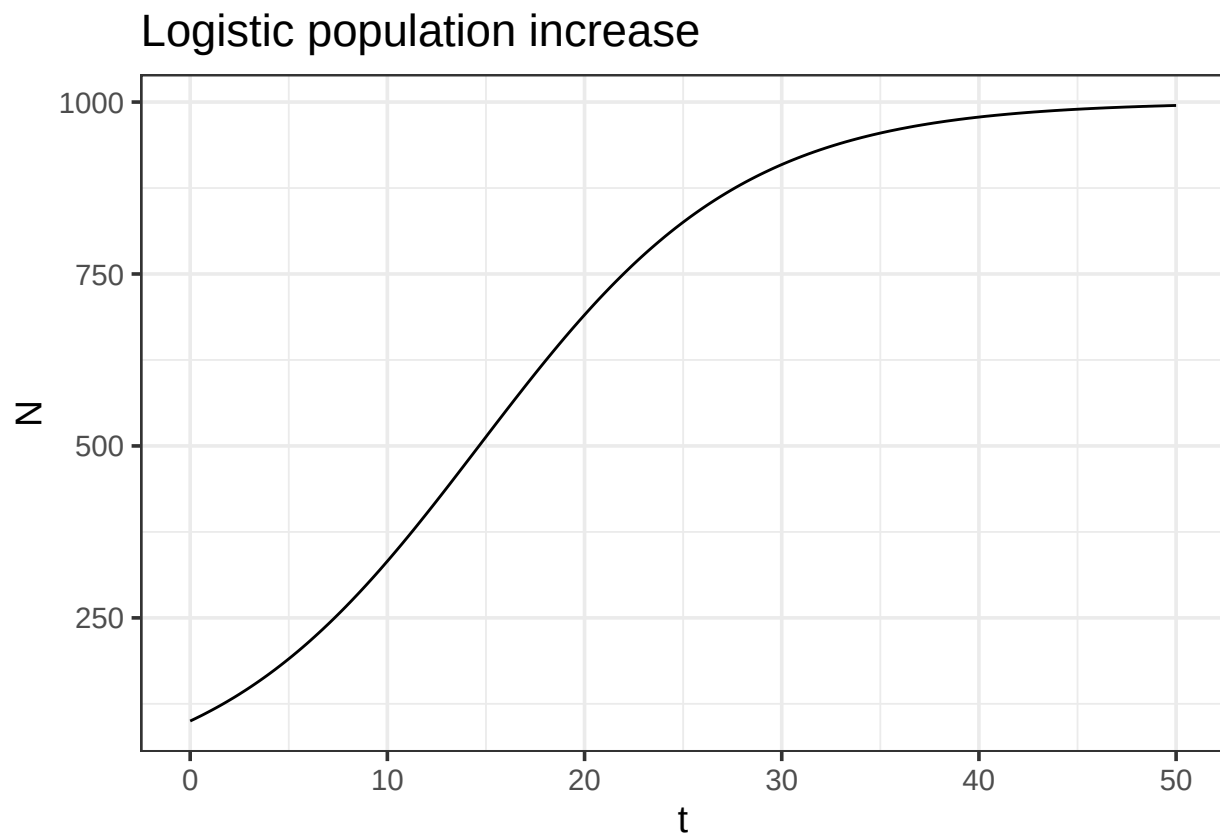
```
logisticN<-function(rm,K,N0,tt) {
  Nt<-K/(1+(K-N0)*exp(-(rm*tt))/N0)
  Nt
}
tt=50
N0=1000
logisticN(rm,K,N0,tt)
```

```
## [1] 1000
```

Now use the function to make a data frame including numbers per year.

```
rm<-0.15
K<-1000
N0<-100
Population<-data.frame(t=seq(0,50,0.1))
Population$N<-logisticN(rm=rm,K=K,N0=N0,tt=Population$t)

ggplot(Population,
       aes(x=t,y=N))+
  geom_line()+
  ggtitle("Logistic population increase")
```



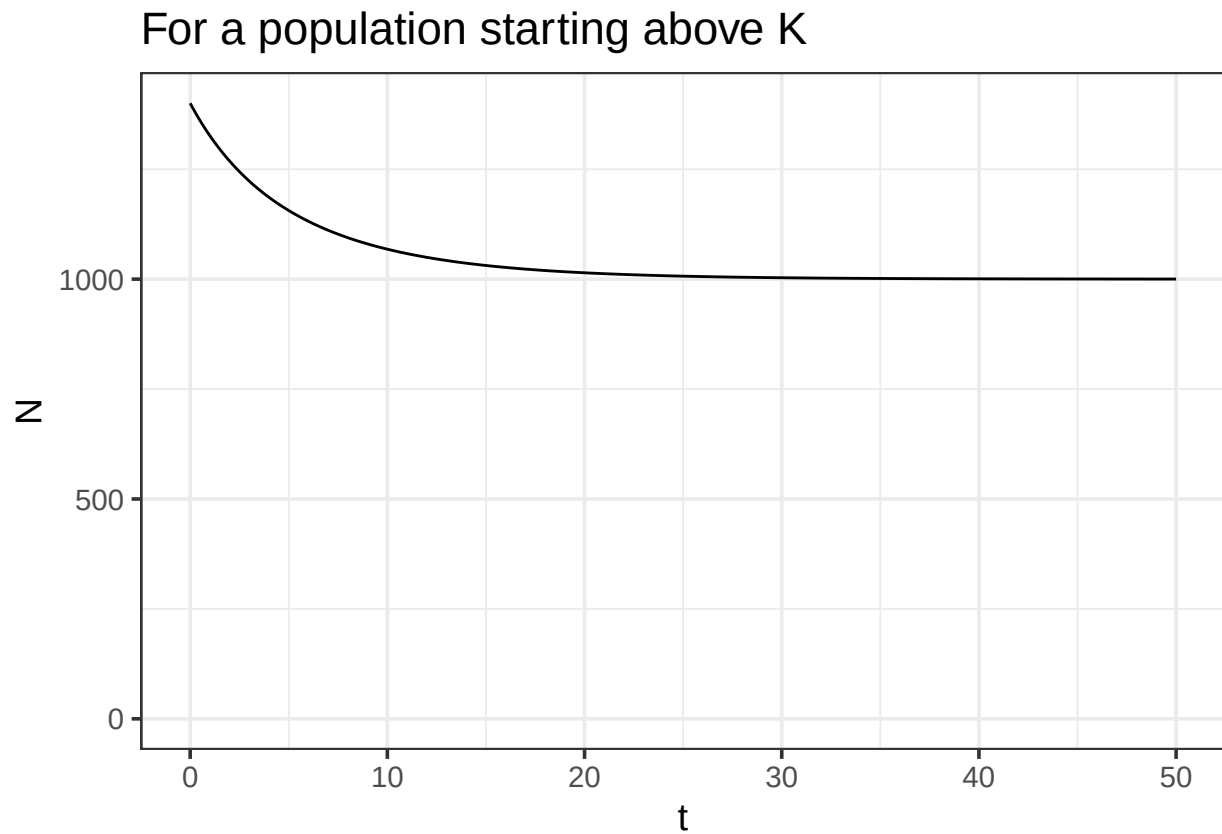
This is logistic population growth, and is a commonly used model for population dynamics for many kinds of species.

If the population starts out above K, it will decline to K

```
rm<-0.15
K<-1000
N0<-1400
Population<-data.frame(t=seq(0,50,0.1))
Population$N<-logisticN(rm=rm,K=K,N0=N0,tt=Population$t)

ggplot(Population,
       aes(x=t,y=N))+
  geom_line()+
  ylim(c(0,N0))+
```

```
ggtitle("For a population starting above K")
```



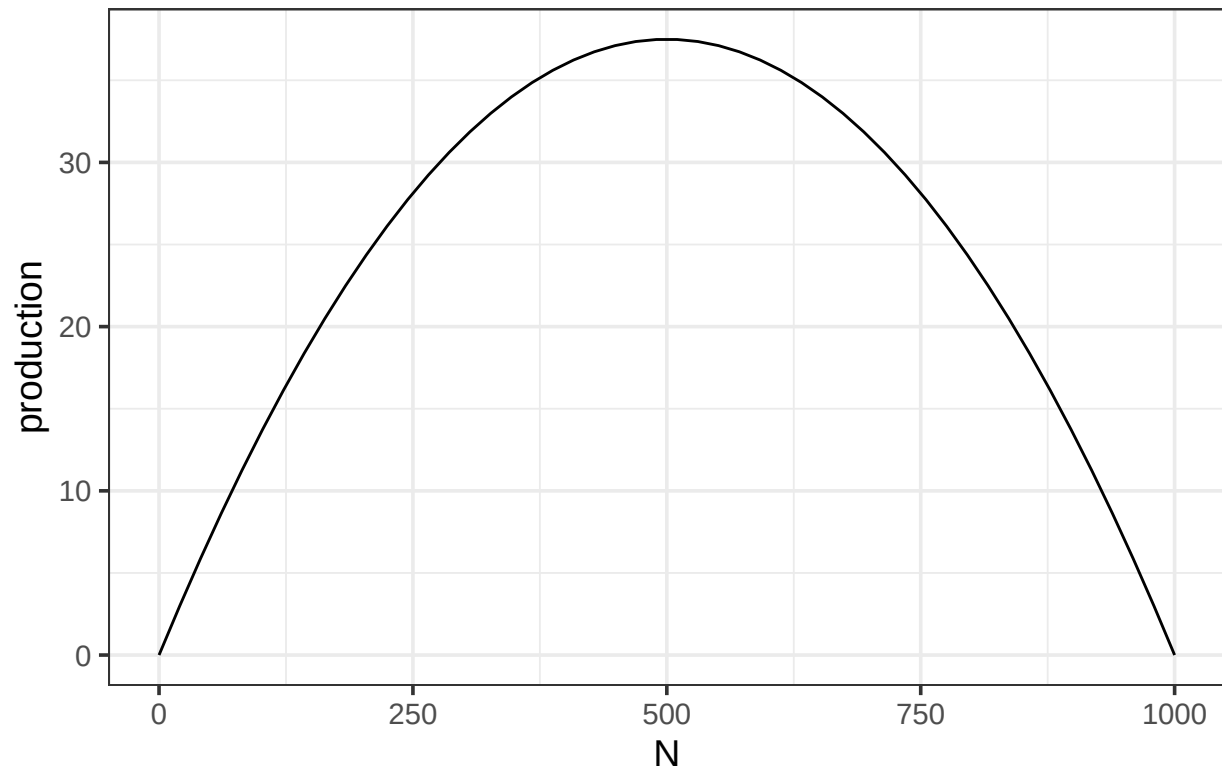
If it starts at K , then it will stay there.

Yield curve and MSY

Make a plot of the increase as a function of abundance.

```
production<-data.frame(N=seq(0,K,length=50))
production$production=rm*(1-production$N/K)*production$N
ggplot(production,aes(x=N,y=production))+
  geom_line()+
  ggtitle("Surplus Production ")
```

Surplus Production



Discrete time version

For an annual time step, this simplifies to

$$N_{t+1} = N_t + r(1/N_t/K)N_t$$

As a function:

```
logisticNdiscrete<-function(rm,K,Nt) {
  Nt+rm*(1-Nt/K)* Nt
}
logisticNdiscrete(rm=0.25,K=1000,Nt=45)
```

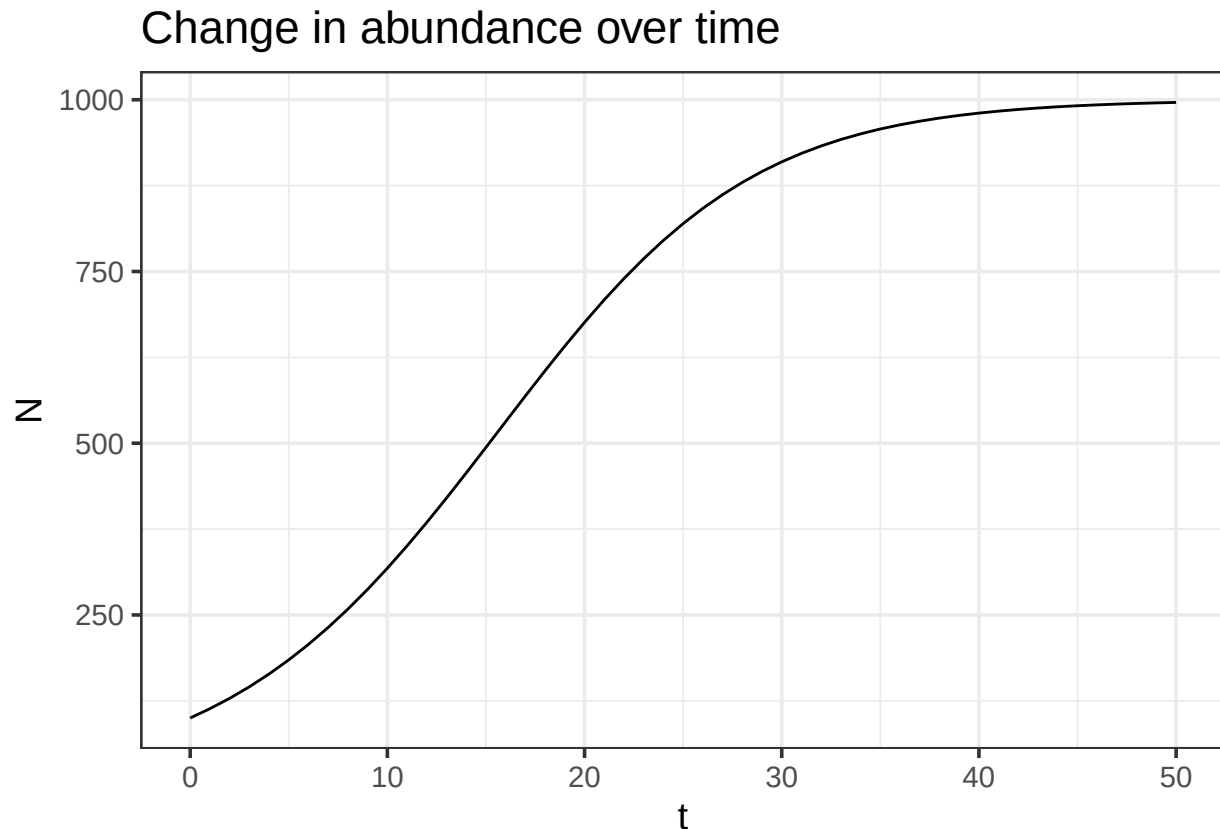
```
## [1] 55.74375
```

To calculate the numbers in each year, start at N_0 , and use the discrete time logistic to calculate the numbers in each following year.

```
rm<-0.15
K<-1000
N0<-100
Population<-data.frame(t=seq(0,50),
  N=NA)
Population$N[1]<-N0
```

```
for(i in 2:nrow(Population)) {
  Population$N[i]<-logisticNdiscrete(rm=rm,K=K,Nt=Population$N[i-1])
}

ggplot(Population,
       aes(x=t,y=N))+
  geom_line()+
  ggtitle("Change in abundance over time")
```



What happens if there is a catastrophic mortality event, like a hurricane?

The discrete time version makes it easy to model. We can write the equation as

$$N_{t+1} = N_t + r(1/N_t/K)N_t - C_t$$

Where C_t is the number of animals killed in the catastrophe. You can see from the equation that the number of animals in time $t + 1$ will be less than the number of animals that are produced by the population $r(1/N_t/K)N_t$. That number is called the surplus production, and it is the amount of new population that can be generated from births minus deaths given the current population.

You may have seen this model before in fisheries, where it is called the Schaefer Surplus Production model, and the C_t stands for the catch in each year. In fisheries, the equation is often calculated in biomass rather than numbers.

For now, let's try an example of a simple mortality event in one year, such as fire or storm. The catastrophe caused 100 extra animals to die in the 10th year, beyond the natural mortality rate. We will put the number killed in a column called C in the Population data


```
Population<-data.frame(t=seq(0,50),
                      C=0,
                      N=NA)
Population$C[Population$t==10]<-90
```

For convenience, add the catch/catastrophe to our function

```
logisticNdiscreteWithC<-function(rm,K,Nt,Ct=0) {
  Nt+rm*(1-Nt/K)* Nt -Ct
}
logisticNdiscreteWithC(rm=0.25,K=1000,Nt=45)
```

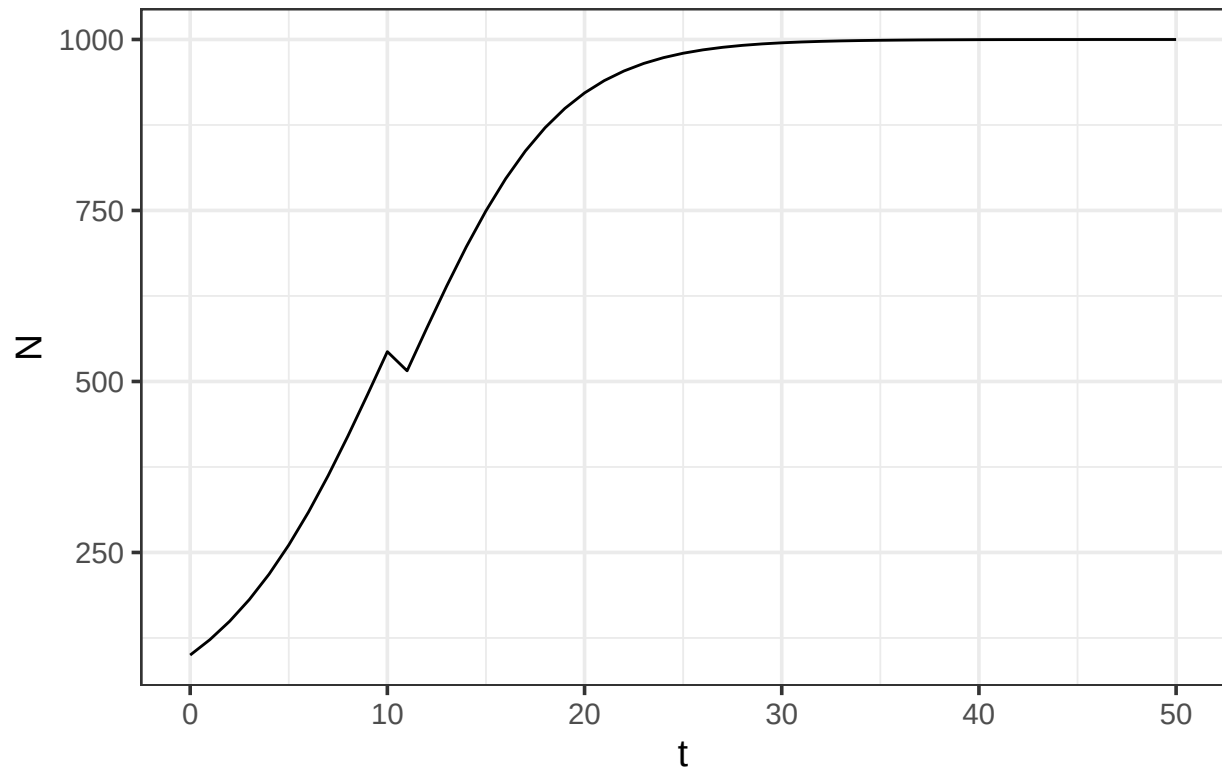
```
## [1] 55.74375
```

Plot with a disaster event

```
rm<-0.25
K<-1000
N0<-100
Population$N[1]<-N0
for(i in 2:nrow(Population)) {
  Population$N[i]<-logisticNdiscreteWithC(rm=rm,K=K,
                                          Nt=Population$N[i-1],
                                          Ct=Population$C[i-1])
}

ggplot(Population,
       aes(x=t,y=N))+
  geom_line()+
  ggtitle("Recovering from a catastrophe" )
```

Recovering from a catastrophe

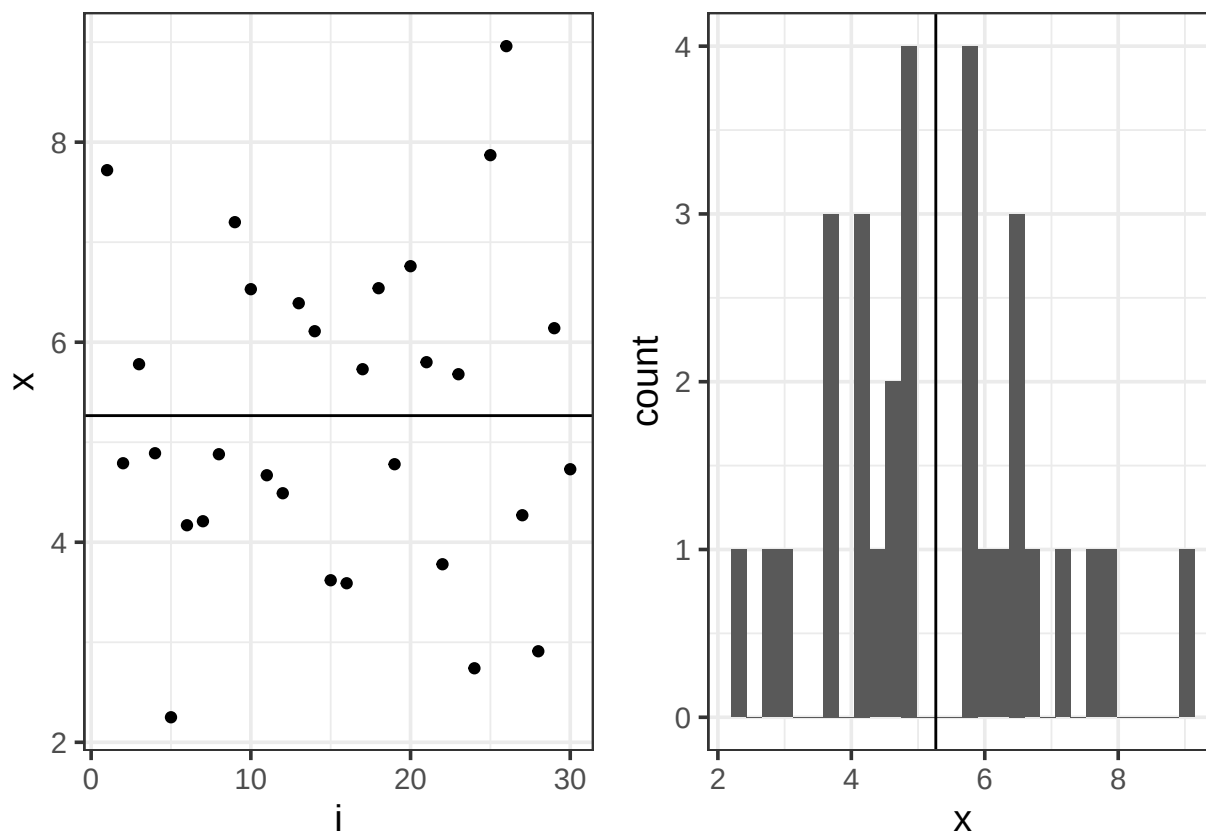


Fitting a model with maximum likelihood

This is some code to fit a very simple model with maximum likelihood so you can see how it works. Then we will fit a surplus production model with likelihood

```
data<-data.frame(i=1:30,x=c(7.72, 4.79, 5.78, 4.89, 2.25, 4.17, 4.21, 4.88, 7.2, 6.53, 4.67, 4.49, 6.39,
3.78, 5.68, 2.74, 7.87, 8.96, 4.27, 2.91, 6.14, 4.73))
g1<-ggplot(data,aes(x=i,y=x))+
  geom_point()+
  geom_hline(yintercept=mean(data$x))
g2<-ggplot(data,aes(x=x))+
  geom_histogram()+
  geom_vline(xintercept=mean(data$x))
gridExtra::grid.arrange(g1,g2,ncol=2)
```

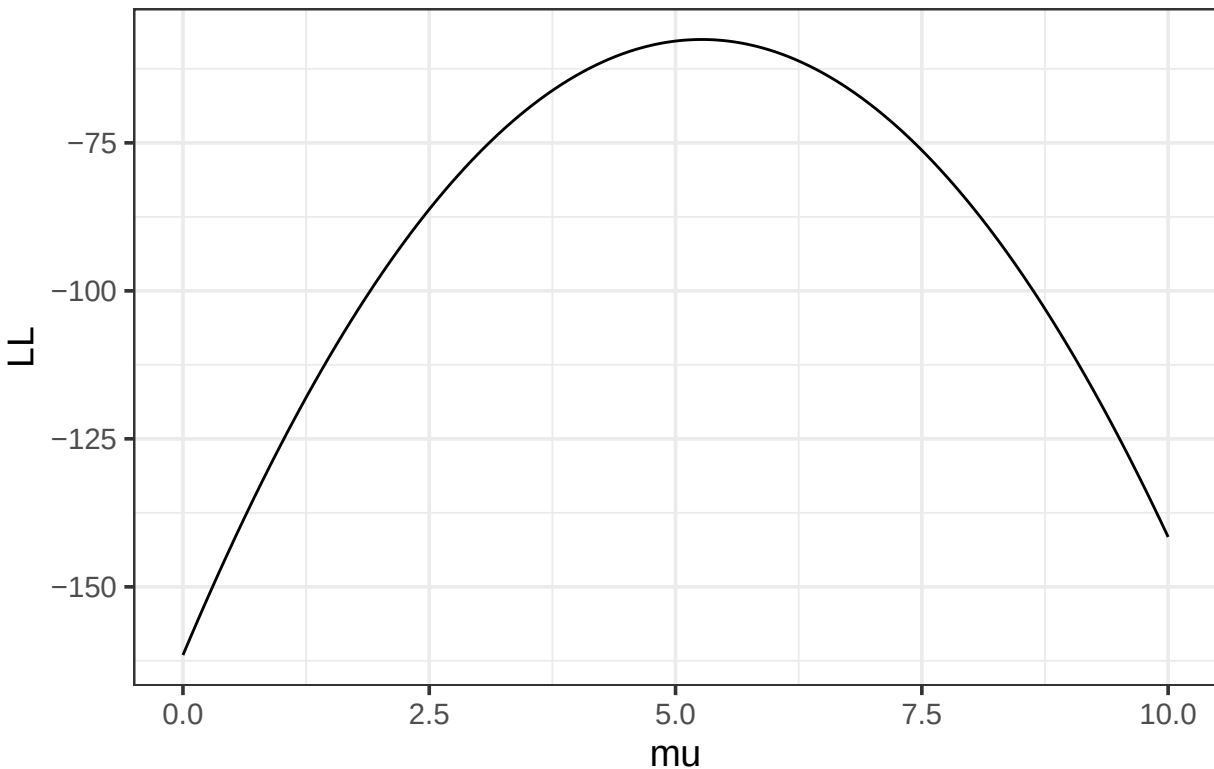
```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```



For simplicity, let's fix the standard deviation and just estimate the mean, using a normal likelihood. First plot the likelihood profile

```
likelihood<-data.frame(mu=seq(0,10,0.01))
for(i in 1:nrow(likelihood)) {
  likelihood$LL[i]=sum(dnorm(data$x,mean=likelihood$mu[i],sd=2,log=TRUE))
}
ggplot(likelihood,aes(x=mu,y=LL))+
  geom_line()+
  ggtitle("Likelihood profile of the mean")
```

Likelihood profile of the mean



```
mean(data$x)
```

```
## [1] 5.266
```

```
likelihood[likelihood$LL==max(likelihood$LL),]
```

```
##      mu      LL  
## 528 5.27 -57.53485
```

Now make a function to minimize to get the MLE (maximum likelihood estimate)

```
getNegLogLike<-function(theta,data) {  
  LL<-dnorm(x=data$x, #data is an input, with a variable called x  
            mean=theta["mu"], #mean is the parameter we want to calculate  
            sd=2,  
            log=TRUE) #fix standard deviation for this example  
  -sum(LL) #return negative total log likelihood  
}  
getNegLogLike(c(mu=5.2),data)
```

```
## [1] 57.55112
```

Use minimizer function optim to find the value of mu that has the lowest -LL, which is the highest LL

```
modFit<-optim(par=c(mu=3),
              fn=getNegLogLike,
              data=data,
              method="L-BFGS-B")
modFit$par
```

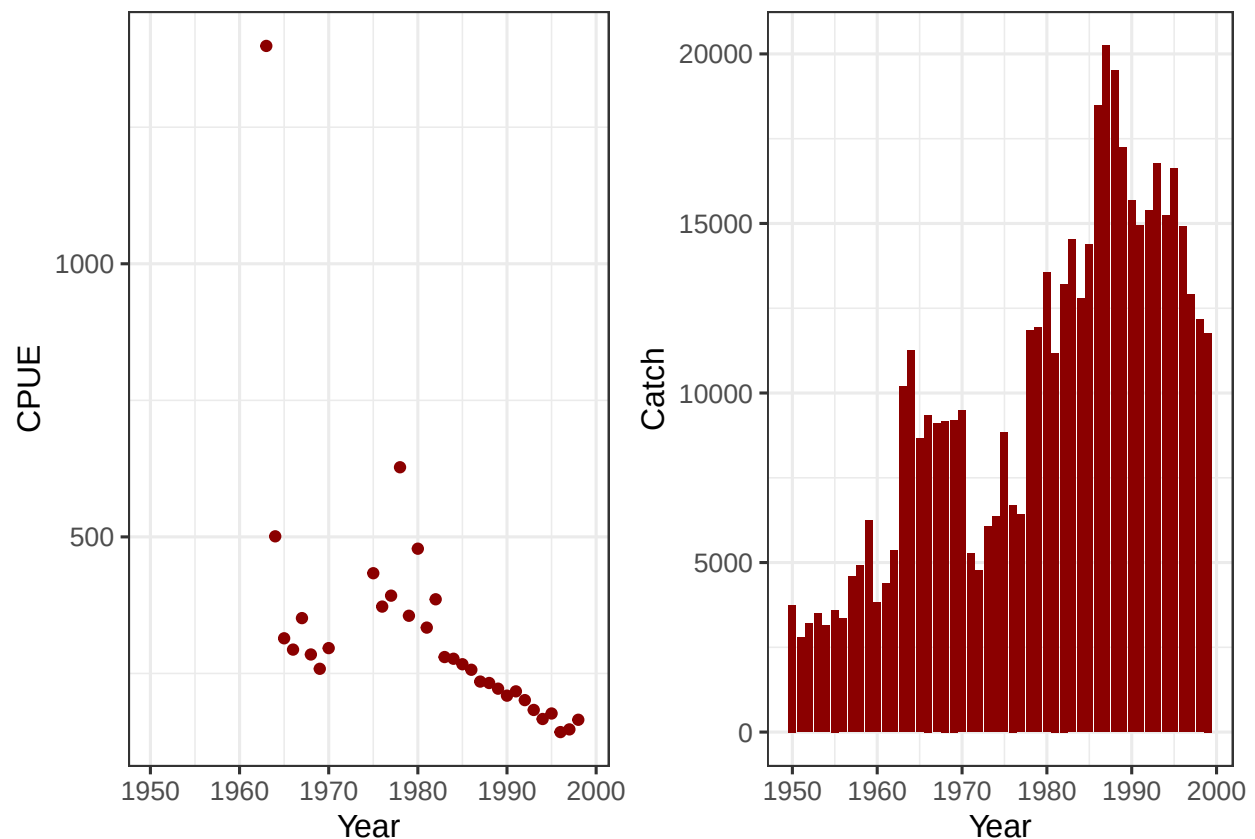
```
##      mu
## 5.266
```

With swordfish surplus production model

Swordfish data

```
swordfish<-read.csv("Swordfish.csv")
swo1<-ggplot(swordfish,aes(x=Year,y=CPUE))+
  geom_point(color="darkred")+theme_bw(base_size=12)
swo2<-ggplot(swordfish,aes(x=Year,y=Catch))+
  geom_col(fill="darkred")+theme_bw(base_size=12)
gridExtra::grid.arrange(swo1,swo2,ncol=2)
```

```
## Warning: Removed 18 rows containing missing values or values outside the scale range
## ('geom_point()').
```



Write function for minimization of -LL

```

getNegLogLike<-function(theta,data) {
  B<-NULL
  LL<-NULL
  logpredI<-NULL
  K<-exp(theta["logK"])
  B[1]<-K
  for(y in 2:nrow(data)) {
    B[y]<-max(c(B[y-1]+theta["r"]*B[y-1]*(1-B[y-1]/K)-
              data$Catch[y-1],0.0001))
  }
  for(y in 1:nrow(data)) {
    logpredI[y]<-log(B[y]*theta["q"])
    LL[y]<-dlnorm(data$CPUE[y],logpredI[y],theta["sigma"],log=TRUE)
  }
  -sum(LL,na.rm=TRUE)
}

getNegLogLike(theta=c(r=0.8,logK=log(90000),q=0.001,sigma=0.3),
               data=swordfish)

```

```
## [1] 562.305
```

Fit the model

```

modFit<-optim(par=c(r=0.8,logK=log(90000),q=0.001,sigma=0.3),
              fn=getNegLogLike,
              data=swordfish,
              lower=c(r=0.01,K=10,q=0.0000001,sigma=0.0001),
              upper=c(r=4,K=1000000,q=1,sigma=10),
              method="L-BFGS-B",
              control=list(trace=TRUE))

```

```

## iter    10 value 186.049896
## final   value 185.185923
## converged

```

```
modFit$par
```

```

##           r           logK           q           sigma
## 0.798782042 11.230539562 0.006235967 0.298333306

```

```
exp(modFit$par["logK"])
```

```

##      logK
## 75398.27

```

Plot the fit

```

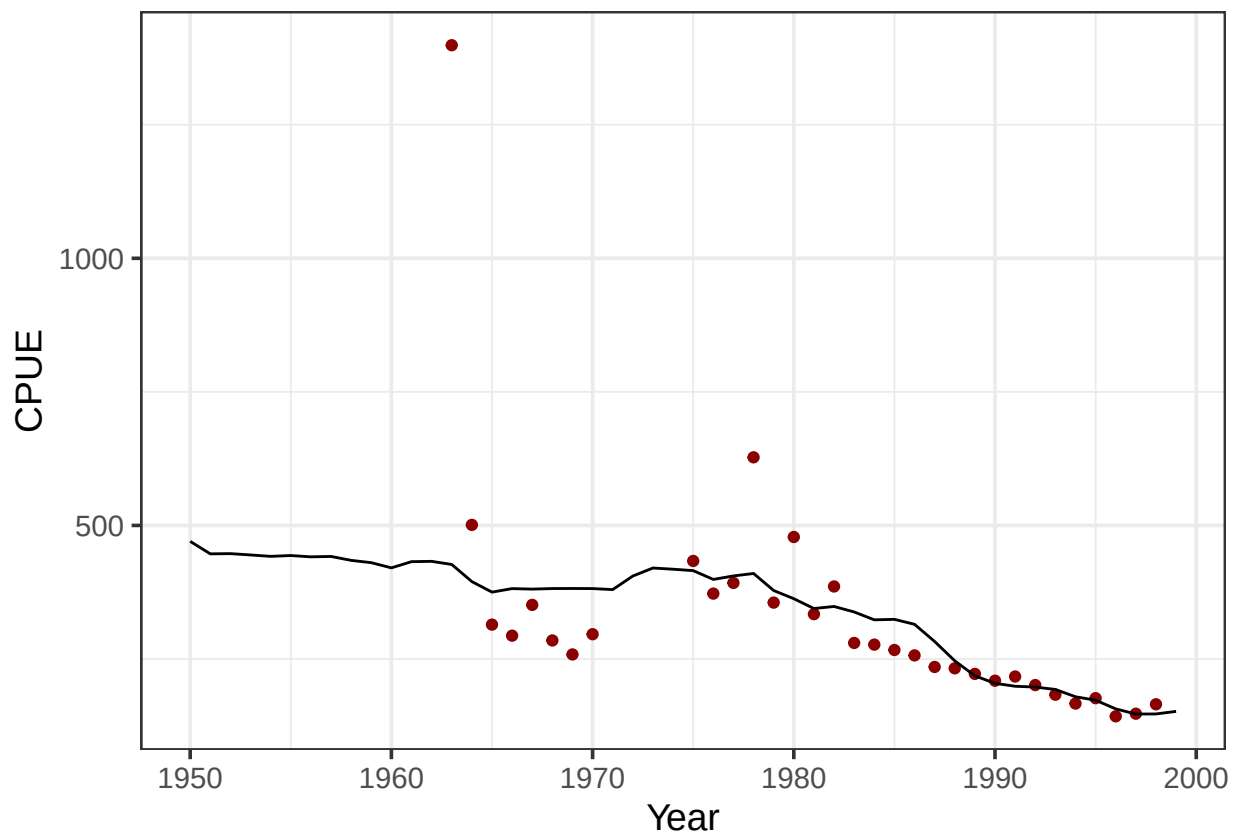
swordfish$Biomass[1]<-exp(modFit$par["logK"])
for(y in 2:nrow(swordfish)) {
  swordfish$Biomass[y]<-logisticNdiscreteWithC(rm=modFit$par["r"],
                                              K=exp(modFit$par["logK"]),
                                              Nt=swordfish$Biomass[y-1],
                                              Ct=swordfish$Catch[y-1])
}
swordfish$Predicted.CPUE<-modFit$par["q"]*swordfish$Biomass
ggplot(swordfish,aes(x=Year,y=CPUE))+
  geom_point(color="darkred")+
  geom_line(aes(y=Predicted.CPUE))

```

```

## Warning: Removed 18 rows containing missing values or values outside the scale range
## ('geom_point()').

```



Run the shiny app and answer all the questions.

<https://elizabeth-babcock.shinyapps.io/ProductionModelDemo2/>